Zeros of functions

Def. $f\colon I_0 = [a_0, b_0] \to \mathbb{R}$ with $f(a_0)\,f(b_0) < 0$. Assume $\exists! \alpha \in I_0$ s.t. $f(\alpha) = 0$.

- Bisection: sequence of intervals $I_k = [a_k, b_k]$, $x_k = \frac{a_k+b_k}{2}$, if $f(x_k) = 0$ we are done, otherwise
  - $I_{k+1} = [a_k, x_k]$ if $f(a_k)\,f(x_k) < 0$
  - $I_{k+1} = [x_k, b_k]$ if $f(b_k)\,f(x_k) < 0$

  If we approximate α by x_k, the error is $|E| < \frac{b_0-a_0}{2^{k+1}}$. Geometric convergence.
- Secant: given $(x_{k-1}, f(x_{k-1}))$ and $(x_k, f(x_k))$, let $g(x)$ be the line through them. Take x_{k+1} where $g(x) = 0$, $x_{k+1} = x_k - f(x_k)\frac{x_{k+1}-x_k}{f(x_{k+1})-f(x_k)}$. Not always convergent, faster than bisection.
- Regula falsi: $I_k = [x_{k-1}, x_k]$, compute x_{k+1} as in the secant method, choose I_{k+1} as in bisection. Convergent if in $I_1 = [x_0, x_1]$, $f(x_0)f(x_1) < 0$. Slower than secant.
- Newton: given $(x_k, f(x_k))$, let $g(x)$ be the line through the point with slope $f'(x_k)$. Take x_{k+1} where $g(x) = 0$, $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$. Not always convergent.

Thm. $f\colon [a,b] \to \mathbb{R}$, class $\mathscr{C}^2$ s.t.:

(i) $f(a)f(b) < 0$

(ii) $f'(x) \neq 0\ \forall x \in [a,b]$

(iii) $f''(x) \geq 0$ (or $f''(x) \leq 0$) $\forall x \in [a,b]$

(iv) $c \in \{a,b\}$ is where $|f'(x)|$ is smaller $\Rightarrow \left|\frac{f(c)}{f'(c)}\right| \leq b-a$

Then $\exists! \alpha \in [a,b]$ such that $f(\alpha) = 0$ and Newton's method converges to α $\forall x_0 \in [a,b]$.

Def. $\{x_k\}_{k\geq 0}$ converging to α has order of convergence at least p if $\exists C > 0$, $N \geq 0$ s.t. $|x_{k+1} - \alpha| \leq C|x_k - \alpha|^p$ $\forall k \geq N$. Equivalently, if $\exists \lim_{k\to\infty} \frac{x_{k+1}-\alpha}{(x_k-\alpha)^p}$ we say it has ord. conv. at least p. If $p = 1$, we require $L < 1$. L is the asymptotic error constant.

Prop. Iteration $x_{k+1} = g(x_k)$, s a fixed point. If $g \in \mathscr{C}^\infty(s-\varepsilon, s+\varepsilon)$, $g^{(j)}(s) = 0$ for $j = 0 \div p-1$ and $g^{(p)}(s) \neq 0$, then $\{x_k\}_{k\geq 0}$ converges with order p.

Obs. (Aitken acceleration) Assume $x_k \to s$. Define $\Delta x_k := x_{k+1} - x_k$, $\Delta^2 x_k := \Delta x_{k+1} - \Delta x_k$. Let $g(x)$ be the line through $(x_k, \Delta x_k)$, $(x_{k+1}, \Delta x_{k+1})$. Take x'_k where $g(x) = 0$, $x'_k = x_k - \frac{(\Delta x_k)^2}{\Delta^2 x_k}$.

Prop. Assume $\lim_{k\to\infty} x_k = s$, $x_k \neq 0$ $\forall k$, and $\exists C$, $|C| < 1$ s.t. $x_{k+1} - s = (C + \delta_k)(x_k - s)$, with $\lim_{k\to\infty} \delta_k = 0$. Then $\{x'_k\}_{k\geq 0}$ is well-defined for k large enough and $\lim_{k\to\infty} \frac{x'_k - s}{x_k - s} = 0 \Rightarrow \{x'_k\}_{k\geq 0}$ converges faster than $\{x_k\}_{k\geq 0}$.

Prop. (Steffensen acceleration) $g\colon [a,b] \to \mathbb{R}$ of class $\mathscr{C}^2$, $s \in [a,b]$ with $s = g(s)$ and $g'(s) \neq 1$. Let $G(x) = x - \frac{(g(x)-x)^2}{g(g(x))-2g(x)+x}$. If $x_{k+1} = g(x_k)$ conv. at least linearly to s, then $y_{k+1} = G(y_k)$ conv. at least quadratically to s. If $x_{k+1} = g(x_k)$ conv. with order $p > 1$, then $y_{k+1} = G(y_k)$ conv. with order $2p-1$.

Obs. Aitken works on the original sequence, Steffensen builds the accelerated sequence.

Thm. (Fixed point) $T \subset \mathbb{R}^n$ closed, $G\colon T \to \mathbb{R}^n$ s.t.:

(i) $G(T) \subset T$ (contractive)

(ii) $\exists L \in (0,1)$ constant s.t. $\|G(x_1) - G(x_2)\| \leq L\|x_1 - x_2\|$ $\forall x_1, x_2 \in T$ (Lipschitz).

Then:

(a) $\exists! s \in I$ s.t. $s = G(x)$ (unique fixed point)

(b) $\forall x_0 \in T$, $\{x_{k+1}\}_{k\geq 0} = \{G(x_k)\}_{k\geq 0}$ converges to s.

(c) $\|x_k - s\| \leq \frac{L^k}{1-L}\|x_1 - x_0\|$, $\|x_k - s\| \leq \frac{L}{1-L}\|x_k - x_{k-1}\|$

Obs. (Newton for nonlinear systems) Let $f\colon \mathbb{R}^n \to \mathbb{R}^n$, given $x_k \in \mathbb{R}^n$, take Taylor $f(x) \approx f(x_k) + Df(x_k)(x - x_k)$. Let $\Delta x_k = x_{k+1} - x_k$, choose x_{k+1} s.t. $f(x) = 0$. Then solve $Df(x_k)\Delta x_k = -f(x_k)$ and $x_{k+1} = x_k + \Delta x_k$.

Obs. If $p(x) \in \mathbb{R}_n[x]$, we write $p(x) = \sum_{k=0}^n a_k x^{n-k}$.

Obs. (Horner's rule) Let $p(x) = \sum_{k=0}^n a_k x^{n-k}$, group it as $p(x) = (\cdots(a_0 x + a_1)x + a_2)x + \cdots a_{n-2})x + a_{n-1})x + a_n$. We want $p(\alpha)$. Define $b_0 = a_0$, $b_i = b_{i-1}\alpha + a_i$, $i = 1 \div n$. Then $p(\alpha) = b_n$.

Obs. (Synthetic division) Horner's rule performs the division of $p(x)$ by $x - \alpha$, $p(x) = (x-\alpha)q(x) + r_p$ where $q(x) = \sum_{k=0}^{n-1} b_k x^{n-1-k}$ and $r_p = b_n = p(\alpha)$.

Obs. (Derivatives at α) Expand p in Taylor around α, $p(x) = p(\alpha) + p'(\alpha)(x-\alpha) + \cdots \frac{p^{(n)}(\alpha)}{n!}(x-\alpha)^n$. Then $q(x) = \frac{p(x)-p(\alpha)}{x-\alpha} = p'(\alpha) + (x-\alpha)\sum_{j=2}^n \frac{p^{(j)}(\alpha)}{j!}(x-\alpha)^{j-2}$. Synthetic division of $q(x) \Rightarrow q(x) = (x-\alpha)\sum_{k=0}^{n-2} c_k x^{n-2-k} + c_{n-1}$ and $q(\alpha) = p'(\alpha) = c_{n-1}$. Iterate:

$$
\begin{array}{llll}
b_0 = a_0 & b_i = b_{i-1}\alpha + a_i & i = 1 \div n & p(\alpha) = b_n \\
c_0 = b_0 & c_i = c_{i-1}\alpha + b_i & i = 1 \div n-1 & p'(\alpha) = c_{n-1} \\
d_0 = c_0 & d_i = d_{i-1}\alpha + c_i & i = 1 \div n-2 & p''(\alpha) = 2!\,d_{n-2} \\
e_0 = d_0 & e_i = e_{i-1}\alpha + d_i & i = 1 \div n-3 & p'''(\alpha) = 3!\,e_{n-3}
\end{array}
$$

Prop. (Laguerre's rule) Let $L > 0$ and perform synthetic division of $p(x)$ by $x - L$, $p(x) = (x-L)\sum_{k=0}^{n-1} b_k x^{n-1-k} + b_n$. If $b_i > 0$ (or $b_i < 0$) $\forall i$, then L is an upper bound for the real roots of $p(x)$.

Prop. (Newton's rule) If $p(x) \in \mathbb{R}_n[x]$ and $L \in \mathbb{R}$ satisfies that $p(L)$, $p'(L), \ldots, p^{(n)}(L)$ are positive (or negative), then L is an upper bound for the real roots of $p(x)$.

Prop. Let $p(x) \in \mathbb{C}_n[x]$ and $z \in \mathbb{C}$ be a root of $p(x)$. Then $|z| \leq \max\left\{1, \sum_{i=1}^n \left|\frac{a_i}{a_0}\right|\right\}$.

Prop. Let $p(x) \in \mathbb{C}_n[x]$ and $z \in \mathbb{C}$ be a root of $p(x)$. Then $|z| \leq 1 + \max\left\{\left|\frac{a_1}{a_0}\right|, \cdots, \left|\frac{a_{n-1}}{a_0}\right|\right\}$.

Obs. Bounds for other real roots of $p(x)$:

- Lower bound for positive roots: $\overline{p}(x) := p\left(\frac{1}{x}\right)$, $x \neq 0$. $\overline{L} > 0$ upper bound for positive real roots of $\overline{p}(x) \Rightarrow \frac{1}{\overline{L}}$ lower bound for positive real roots of $p(x)$.
- Lower bound for negative roots: $\overline{p}(x) := p(-x)$. $\overline{L} > 0$ upper bound for real roots of $\overline{p}(x) \Rightarrow -\overline{L}$ lower bound for negative real roots of $p(x)$.
- Upper bound for negative roots: $\overline{p}(x) := p\left(-\frac{1}{x}\right)$. $\overline{L} > 0$ upper bound for real roots of $\overline{p}(x) \Rightarrow -\frac{1}{\overline{L}}$ upper bound for negative real roots of $p(x)$.

Polynomial interpolation of functions

Grid (x_k, f_k), $k = 0 \div n$ of $n+1$ points, $x_i \neq x_j$ $\forall i \neq j$

Thm. (Existence and uniqueness) $\exists!\, p_n(x)$ polynomial of degree $\leq n$ that interpolates (x_k, f_k), i.e. $p_n(x_k) = f_k$, $k = 0 \div n$. Idea: $p_n(x) = \sum_{i=0}^n \left(C_i \prod_{j=0}^{i-1}(x - x_j)\right)$.

Def. $\langle x_0, \ldots, x_n\rangle := (\min\{x_0, \ldots, x_n\}, \max\{x_0, \ldots, x_n\})$.

Thm. (Error formula) If $f \in \mathscr{C}^{n+1}(a,b)$, $x_k \in (a,b)$ $k = 0 \div n$ and $p_n(x)$ is the interpolating polynomial, then $f(x) - p_n(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x - x_0)\ldots(x - x_n)$ with $\xi(x) \in \langle x_0, \ldots, x_n, x\rangle$.

Def. (Lagrange pol.) $\ell_k(x) = \prod_{i\neq k}(x - x_i) / \prod_{i\neq k}(x_k - x_i)$, where $\ell_k(x_j) = \delta_{jk}$.

Obs. (Lagrange method) If $p_n(x)$ is the interpolating polynomial of (x_k, f_k) with $k = 0 \div n$, then $p_n(x) = \sum_{k=0}^n f_k \ell_k(x)$.

Def. Grid (x_k, f_k) with $k = 0 \div n$, define $f[x_i] := f_i$ and $f[x_i, \ldots, x_{i+j+1}] = \frac{f[x_{i+1},\ldots,x_{i+j},x_{i+j+1}] - f[x_i,x_{i+1},\ldots,x_{i+j}]}{x_{i+j+1} - x_i}$.

Obs. (Newton method) If $p_n(x)$ is the interpolating polynomial of (x_k, f_k) with $k = 0 \div n$, then $C_i = f[x_0, \ldots, x_i]$ $i = 0 \ldots n$ and $p_n(x) = \sum_{i=0}^n \left(C_i \prod_{j=0}^{i-1}(x - x_j)\right)$.

Prop. $\forall \sigma \in S_{j+1}$ permutation of the symmetric group of degree $j+1$, $f[x_0, \ldots, x_j] = f[x_{\sigma(0)}, \ldots, x_{\sigma(j)}]$.

Def. Grid (x_k, f_k) $k = 0 \div n$, define $\Delta^0 f(x_k) = f(x_k)$, $\Delta f(x_k) = f(x_{k+1}) - f(x_k)$, $\Delta(f(x_i) - f(x_k)) = \Delta f(x_i) - \Delta f(x_k)$, $\ldots$, $\Delta^n f(x_k) = \Delta^{n-1}(\Delta f(x_k))$.

Prop. If $h \in \mathbb{R}$ and we have the equally spaced grid $\{x_0,\ x_1 = x_0 + h,\ \ldots, x_n = x_0 + nh\}$, then $f[x_0, \ldots, x_k] = \frac{1}{k!h^k}\Delta^k f(x_0)$.

Inverse interpolation

We know (x_k, f_k) $k = 0 \div n$, we want to solve $f(x) = c$. Assume f invertible and compute $x = g(c)$, $g = f^{-1}$. Interpolate $(y_i, g_i) = (f_i, x_i)$ $i = 0 \div n$ by an interpolating polynomial $q_n(y)$ and take $x_{\text{approx}} = q_n(c)$.From the error formula,

$$
\max_{x\in[a,b]} |f(x) - p_n(x)| \leq
$$
$$
\leq \max_{x\in[a,b]} \left\{\frac{|f^{(n+1)}(x)|}{(n+1)!}\right\} \cdot \max_{x\in[a,b]} \left|\prod_{i=0}^n (x - x_i)\right|
$$

We want to minimize $\max_{x\in[a,b]} |x - x_0| \cdots |x - x_n|$.

Thm. The best choice of $y_0, \dots, y_n \in [-1,1]$ to minimize $\max_{y\in[a,b]} |y-y_0|\cdots|y-y_n|$ is given by the roots of the Chebyshev polynomials of degree $n+1$ and equals $\frac{1}{2^n}$.
Def. Chebyshev pol. degree n: $T_n(x) = \cos(n \arccos(x))$.
Prop. For Chebyshev polynomials:

(i) $T_0(x) = 1$, $T_1(x) = x$, $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$
(ii) Coefficient of x^n in $T_n(x)$ is 2^{n-1}
(iii) $T_n(x)$, $n \geq 1$ has n zeros in $[-1,1]$, of the form $x_k = \cos\left(\frac{2k+1}{2}\frac{\pi}{2}\right)$ $k = 0 \div n-1$
(iv) $T_n(x)$, $n \geq 1$ has $n+1$ extrema in $[-1,1]$, of the form $\overline{x}_k = \cos\left(\frac{k\pi}{n}\right)$ $k = 0 \div n$, $T_n(\overline{x}_k) = (-1)^k$.

Thm. Let $P_{n+1} = \sum_{i=0}^{n+1} a_i x^i$ be any monic polynomial of degree $n+1$ and $m = \max_{x\in[-1,1]} |P_{n+1}(x)|$. Then $\frac{T_{n+1}(x)}{2^n}$ is monic of degree $n+1$ and minimizes m, and $\min_{P_{n+1}(x)} m = \frac{1}{2^n}$.

Hermite interpolation

Goal: find a polynomial matching $f(x)$ and $f'(x)$ on the grid (x_k, f_k), (x_k, f'_k) $k = 0 \div m$.
Prop. The unique polynomial interpolating $f(x)$ and $f'(x)$ at (x_k, f_k), (x_k, f'_k) $k = 0 \div m$ is the Hermite polynomial of degree $2m+1$, $H_{2m+1}(x) = \sum_{i=0}^m f_i\phi_i(x) + \sum_{i=0}^m f_i\psi_i(x)$ where:

(i) $\phi_i(x) = [1 - 2\ell'_i(x_i)(x - x_i)]\,\ell_i^2(x)$
(ii) $\psi_i(x) = (x - x_i)\ell_i^2(x)$

with $\ell_i(x)$ the Lagrange interpolating polynomial of degree m.
Prop. (Error formula) $I \subset \mathbb{R}$ and $f \in \mathscr{C}^{2m+2}(I)$ with $x_k \in I$ for $k = 0 \div m$. $f(x) - H_{2m+1}(x) = \frac{f^{(2m+2)}(\xi(x))}{(2m+2)!}(x - x_0)^2 \cdots (x - x_m)^2$ $\forall x \in I$ with $\xi(x) \in \langle x_0, \dots, x_m, x\rangle$.
Def. (Generalized divided differences) Given x_i and f, $\lim_{x\to x_i} f[x_i, x] = \lim_{x\to x_i} \frac{f(x) - f(x_i)}{x - x_i} = f'(x_i) = f[x_i, x_i]$.
Prop. Computation scheme for $H_{2m+1}(x)$:

$$H_{2m+1}(x) = f_0 + f[x_0,x_0](x-x_0) + f[x_0,x_0,x_1](x-x_0)^2 + \\ + f[x_0,x_0,x_1,x_1](x-x_0)^2(x-x_1) + \dots \\ + f[x_0,x_0,\dots,x_m,x_m](x-x_0)^2\cdots(x-x_{m-1})^2(x-x_m)$$

Spline interpolation

Set of points $x_0 < x_1 < \dots < x_n$ and $f_k = f(x_k)$ $k = 0 \div n$.
Def. A spline $s(x)$ of degree p interpolating f at the nodes x_i $i = 0 \div n$ satisfies:

(i) Interpolates: $s(x_i) = f_i$ $i = 0 \div n$
(ii) $\forall [x_i, x_{i+1}]$ $i = 0 \div n-1$, $s(x)$ is a polynomial of degree p
(iii) $s(x) \in \mathscr{C}^{p-1}([x_0, x_n])$

In practice, natural cubic splines are used ($s''(x_0) = s''(x_n) = 0$).

Obs. We build natural cubic splines. Define $s_i(x) = s(x)|_{[x_i,x_{i+1}]}$ $i = 0 \div n-1$. We need $4n$ coefficients.

$$\left.\begin{aligned} s_i(x_i) &= f_i, & i &= 0 \div n-1 \\ s_{n-1}(x_n) &= f_n \\ s_i(x_{i+1}) &= s_{i+1}(x_{i+1}) & i &= 0 \div n-2 \\ s'_i(x_{i+1}) &= s'_{i+1}(x_{i+1}) & i &= 0 \div n-2 \\ s''_i(x_{i+1}) &= s''_{i+1}(x_{i+1}) & i &= 0 \div n-2 \end{aligned}\right\} 4n-2 \text{ conditions}$$

Impose $s''(x_0) = s''(x_n) = 0$ (natural) $\Rightarrow 4n$ conditions. Define: $h_i = x_{i+1} - x_i$ $i = 0 \div n-1$, $M_i = s''(x_i)$ $i = 0 \div n$. We have $s''_i(x) = M_i + \frac{M_{i+1} - M_i}{h_i}(x - x_i)$ $i = 0 \div n-1$.
Integrate twice between x_i and x: $s_i(x) = M_i\frac{(x-x_i)^2}{2} + \frac{M_{i+1}-M_i}{h_i}\frac{(x-x_i)^3}{6} + B_i(x - x_i) + A_i$. Imposing $s_i(x_i) = f_i = A_i$
$A_i = f_i = s_i(x_i)$ $i = 0 \div n$

$$B_i = \frac{f_{i+1} - f_i}{h_i} - (M_{i+1} - M_i)\frac{h_i}{6} - \frac{M_i h_i}{2} \quad i = 0 \div n-1$$

$$h_i M_i + 2(h_i + h_{i+1})M_{i+1} + h_{i+1}M_{i+2} = 6d_{i+1} \quad i = 0 \div n-2$$

$$d_{i+1} = \frac{f_{i+2} - f_{i+1}}{h_{i+1}} - \frac{f_{i+1} - f_i}{h_i} \quad i = 0 \div n-2$$

$M_0 = M_n = 0$
The d_i are ordinary differences. We have the tridiagonal system $T[M_1 \cdots M_{n-1}]' = [d_1 \cdots d_{n-1}]'$ where

$$T = \begin{bmatrix} 2(h_0+h_1) & h_1 & 0 & & \\ h_1 & 2(h_1+h_2) & h_2 & & \\ 0 & h_2 & 2(h_2+h_3) & h_3 & \\ \vdots & \vdots & \ddots & \ddots & \ddots \end{bmatrix}$$

Compute M_i, then B_i and $s_i(x)$.

Trigonometric interpolation

Interpolate a 2π-periodic function $f\colon [0, 2\pi] \to \mathbb{R}$ by a trigonometric interpolating polynomial $\hat{f}$ of f at $x_j = \frac{2\pi}{n+1}j$ $j = 0 \div n$. We use $e^{ikx} = \cos(kx) + i\sin(kx)$. The interpolating polynomial is

$$\hat{f}(x) = \frac{a_0}{2} + \sum_{k=1}^{M+\mu}(a_k\cos(kx) + b_k\sin(kx)) = \sum_{k=-M-\mu}^{M+\mu} c_k e^{ikx}$$

If n is even, $\mu = 0$, $M = \frac{n}{2}$. If n is odd, $\mu = 1$, $M = \frac{n-1}{2}$ and we impose $c_{M+1} = c_{-(M+1)}$ ($\Rightarrow a_{M+1} = 2c_{M+1}$, $b_{M+1} = 0$).
Coefficient computation:

$$c_l = \frac{1}{n+1}\sum_{j=0}^{n} f(x_k)e^{-ijhl}$$

$$a_k = c_k + c_{-k} = \frac{2}{n+1}\sum_{j=0}^{n} f(x_k)\cos(jhk)$$

$$b_k = i(c_k - c_{-k}) = \frac{2}{n+1}\sum_{j=0}^{n} f(x_k)\sin(jhk)$$

$$a_{M+1} = 2c_{M+1} = \frac{2}{n+1}\sum_{j=0}^{n} f(x_j)e^{-ijh(M+1)} \qquad b_{M+1} = 0$$

Function approximation

Given $\varphi_0(x), \dots, \varphi_n(x)$ l.i. on I, $\mathcal{F}_n = \{f_n = a_0\varphi_0(x) + \dots + a_n\varphi_n(x) \mid a_0, \dots, a_n \in \mathbb{R}\}$. Given f on I, we want $f_n^* \in \mathcal{F}_n$ s.t. $\|f - f_n^*\|_2 = \min_{f_n\in\mathcal{F}_n}\|f - f_n\|_2$. $\|f\|$.
Thm. Given I and l.i. $\{\varphi_i(x)\}_{i=0,\dots,n}$, $\exists! f_n^* = \sum_{k=0}^n a_k^*\varphi_k(x)$ minimizing $\|f - f_n\|_2$ in $\mathcal{F}_n$.
Obs. From LA we know f_n^* is characterized by $\langle f - f_n^*, \varphi_i\rangle = 0$.
From $\{\varphi_j\}_{j=0,\dots,n}$ we obtain orthogonal $\{\psi_j\}_{j=0,\dots,n}$ (Gram-Schmidt). Setting up the system

$$(\varphi_0(x), \dots, \varphi_n(x)) = (\psi_0(x), \dots, \psi_n(x))\begin{pmatrix} r_{00} & r_{01} & \dots & r_{0n} \\ & r_{11} & \dots & r_{1n} \\ & & \ddots & \vdots \\ & & & r_{nn} \end{pmatrix}$$

with $r_{ij} = \frac{\langle\varphi_j,\psi_i\rangle}{d_i}$, where $d_i = \langle\psi_j,\psi_j\rangle$. We get $\psi_0(x) = \varphi_0(x)$, $\psi_j(x) = \varphi_j(x) - \sum_{i=0}^{j-1}\frac{\langle\varphi_j,\psi_i\rangle}{\langle\psi_i,\psi_i\rangle}\psi_i(x)$, $j = 1 \div n$.
Obs. Procedure: from $\{\varphi_j\}$ compute $\{\psi_j\}$, compute $\langle\psi_i,\psi_i\rangle$, solve the normal-eqs. LS system; now diagonal $c_i^* = \frac{\langle\psi_i,f\rangle}{\langle\psi_i,\psi_i\rangle}$.
$f_n^* = c_0^*\psi_0(x) + \dots + c_n^*\psi_n(x)$.
Prop. If $\psi_0(x), \dots, \psi_n(x)$ are orthogonal, $f_n^*(x) = \sum_{j=0}^n c_j^*\psi_j(x)$ is the least-squares function, and $\|f - f_n^*\|^2 = \|f\|^2 - \|f_n^*\|^2$.
Obs. The normal-eqs. LS system can be ill-conditioned. We want a diagonal LS system.

Polynomial approximation case: orthogonal polynomials

Prop. We have:

- $\langle xu, v\rangle = \langle u, xv\rangle$ $\forall u, v \in \mathcal{P}_n$.
- Given $\psi_0, \dots, \psi_i$ with degrees $\{\psi_j(x)\} = j$, orthogonal $\{\psi_j\}_j$ ($\Rightarrow$ l.i.), any polynomial p_i of degree i can be written uniquely as $p_i(x) = c_0\psi_0(x) + \dots + c_i\psi_i(x)$.
- $\langle\psi_j, p_i\rangle = 0$ $\forall p_i(x)$ s.t. $i < j$.

Prop. Least-squares approx. solution given by $p_n^*(x) = \sum_{j=0}^n c_j^*\psi_j(x)$ where c_j^* is the solution of the normal-eqs. LS system (now diagonal) $c_j^* = \frac{\langle\psi_j,f\rangle}{\langle\psi_j,\psi_j\rangle}$, $j = 0 \div n$. Orthogonal polynomials $\psi_j(x)$ are given by the recurrence

$$\text{(RAPO)}\begin{cases} \psi_0(x) = A_0 & (\psi_{-1}(x) = 0) \\ \psi_{j+1}(x) = \alpha_j(x - \beta_j)\psi_j(x) - \gamma_j\psi_{j-1}(x), & j \geq 0. \end{cases}$$

with

$$\begin{cases} \alpha_j = \frac{A_{j+1}}{A_j}, & j \geq 0 \\ \beta_j = \frac{\langle\psi_j, x\psi_j\rangle}{\langle\psi_j,\psi_j\rangle} = \frac{\langle\psi_j, x\psi_j\rangle}{d_j}, & j \geq 0 \\ \gamma_j = \frac{\alpha_j}{\alpha_{j-1}}\frac{\langle\psi_j,\psi_j\rangle}{\langle\psi_{j-1},\psi_{j-1}\rangle} = \frac{\alpha_j d_j}{\alpha_{j-1}d_{j-1}}, & j \geq 1 \end{cases}$$

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